Beta Blocker Toxicity

Section 1: Case Summary

Scenario Title:	
Keywords:	Beta blocker overdose, high-dose insulin, glucagon
Brief Description of Case:	A 44-year-old male presents to the emergency department following the ingestion of an entire bottle of metoprolol. Decontamination strategies should be utilized alongside consultation with poison control. Patient clinically deteriorates as the drug reaches peak effects, requiring IVF, atropine, glucagon, multi-dose vasopressors, high dose insulin, and a discussion around potential salvage therapies.

Goals and Objectives	
Educational Goal:	To allow learners to identify and treat a severe beta blocker overdose requiring maximal medical therapy, including infrequently utilized therapies.
Objectives: (Medical and CRM)	**Medical** 1. Identify beta blocker toxicity 2. Provide decontamination and initial supportive care 3. Initiate appropriate management of severe beta blocker toxicity **CRM** 1. Mobilize resources to complete simultaneous investigation and treatment of an unstable patient including poison control and relevant consultants 2. Anticipate and plan for a potentially critically unwell patient
EPAs Assessed:	**ME 2.4** Initiate medical treatment of the patient with an overdose, toxic ingestion or exposure, including specific antidote therapy.

Learners, Setting and Personnel				
Target Learners:	☐ Junior Learners	☒ Senior Learners	☒ Staff	
	☒ Physicians	☒ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:			
Location:	☐ Sim Lab	☒ In Situ	☐ Other:	
Recommended Number of Facilitators:	Instructors: 1-2			
	Confederates: 0-1			
	Sim Techs: 1			

Scenario Development	
Date of Development:	04/10/2020
Scenario Developer(s):	Donovan MacDonald & Daniel Ovakim
Affiliations/Institutions(s):	University of British Columbia, Departments of Emergency & Critical Care Medicine
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Last Revision Date:	N/A
Revised By:	N/A
Version Number:	1

Section 2A: Initial Patient Information

<table>
<tr><th colspan="6">A. Patient Chart</th></tr>
<tr><td colspan="3">Patient Name: Hunter Campbell</td><td>Age: 44</td><td>Gender: Male</td><td>Weight: 70 kg</td></tr>
<tr><td colspan="6">Presenting complaint: Intentional overdose ingestion – no specific complaints at triage.</td></tr>
<tr><td>Temp: 36.7</td><td>HR: 90</td><td>BP: 120/75</td><td>RR: 15</td><td>O_2Sat: 97%</td><td>FiO_2: RA</td></tr>
<tr><td colspan="3">Cap glucose: 6.5 mmol/L</td><td colspan="3">GCS: 15 (E4 V5 M6)</td></tr>
<tr><td colspan="6">Triage note: 44-year-old male with an intentional overdose ingestion. States fight with partner. Remorseful and concerned he will become unwell.</td></tr>
<tr><td colspan="6">Allergies: NKDA</td></tr>
<tr><td colspan="3">Past Medical History:
Supraventricular tachycardia</td><td colspan="3">Current Medications:
Metoprolol 50mg p.o. BID</td></tr>
</table>

Section 2B: Extra Patient Information

<table>
<tr><th colspan="2">A. Further History</th></tr>
<tr><td colspan="2">Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin’s voice, confederate, SP, etc.)?

Further HPI to be provided if the learner asks:
The patient had just filled his prescription at the pharmacy adjacent to the hospital, and impulsively took the entire bottle after leaving work. The estimated ingestion is of 100 x 50 mg tablets (5000 mg) approximately 45 minutes prior to presentation.</td></tr>
<tr><th colspan="2">B. Physical Exam</th></tr>
<tr><td colspan="2">List any pertinent positive and negative findings</td></tr>
<tr><td>Cardio: S1, S2 without murmurs, rubs or extra heart sounds. Palpable peripheral pulses x4.</td><td>Neuro: Alert and oriented, appears anxious with a normal screening neurologic exam.</td></tr>
<tr><td>Resp: Breath sounds equal bilaterally, no adventitious sounds.</td><td>Head & Neck: Unremarkable.</td></tr>
<tr><td>Abdo: Soft, non-tender</td><td>MSK/skin: Normal</td></tr>
<tr><td colspan="2">Other: No distress at present but anxious he will get worse.</td></tr>
</table>

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (Adult)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Gloves Stethoscope Defibrillator NIBP cuff / Pulse oximeter Cardiac monitor IV Bags/Lines IV Push Medications Nasal Prongs Non-Rebreather Mask Bag Valve Mask Laryngoscope ET/NG/OG Tubes
C. Required Medications
Calcium gluconate / Calcium chloride Glucagon Insulin /Dextrose Charcoal Antiemetics Atropine Norepinephrine Epinephrine Vasopressin Dobutamine Milrinone Phenylephrine Lipid emulsion 20%
D. Moulage
Young adult male in regular street wear. Consider activated charcoal moulage if treated with the same.
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format. A/B: Normal C: Peripheral pulses become weak or non-palpable with worsening hypotension D: GCS fluctuates with worsening hypotension E: Normal

Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	*Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)*
Nurse	Communicate poison control recommendations as needed to facilitate appropriate medical management.

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State (45 mins post ingestion) Rhythm: Sinus HR: 90 BP: 120/75 RR: 15 O_2SAT: 97% T: 36.7ºC GCS: 15	*Alert, but anxious.*	<u>Expected Learner Actions</u> ☐ History / Physical ☐ Monitors ☐ Establish IV access ☐ Order ECG / blood work ☐ Confirm quantity / time of ingestion and inquire regarding potential co-ingestions ☐ Recognize severity of overdose ingestion and early consideration of decontamination	<u>Modifiers</u> *Changes to patient condition based on learner action* - If they call poison control, will be prompted to talk to them over the phone by nurse who will get back to them. - Requests ECG → ECG #1 - Administration of activated charcoal (moulage) <u>Triggers</u> *For progression to next state* - All actions complete or **5 min** into case → **2. Initial Resuscitation**	*Note: Time frame for this case should be as listed in triggers and not as listed in time post ingestion.
2. Initial Resuscitation (60 mins post ingestion) Rhythm: Sinus HR: 45 BP: 90/45 RR: 18 O_2SAT: 98% T: 36.7ºC GCS: 15	*Alert, vague symptoms of pre-syncope, lightheadedness*	<u>Expected Learner Actions</u> ☐ Recognize early hemodynamic instability ☐ Establish second IV ☐ IVF bolus of crystalloid (2L) ☐ Trial Atropine 0.5-1mg IV ☐ Targeted treatment with IV Glucagon 1-3mg IV/Calcium gluconate 3-6amps IV ☐ Review information provided by poison control and anticipate potential for further decline	<u>Modifiers</u> - Supportive treatment with IVF → Transient rise; consider 3rd litre - Glucagon → Trigger nausea and vomiting requiring antiemetics; no change in hemodynamics - Atropine → No effect <u>Triggers</u> - All actions complete or **10 min** into case → **2. Refractory hypotension**	

3. Refractory hypotension (120 mins post ingestion) Rhythm: Sinus HR: 50 BP: 88/42 RR: 18 O_2SAT: 96% T: 36.7ºC GCS: 15	*Drowsy and unwell*	<u>Expected Learner Actions</u> ☐ Recognize refractory hemodynamics ☐ Escalate to vasopressors (starting doses with no max dose in beta blocker overdose) - NEpi 0.1 mcg/kg/min - Epi 0.1 mcg/kg/min (start when NEpi at 0.4) - Vaso 2.4 units/hour (fixed) ☐ Consider central venous access ☐ Initiate high-dose insulin at 1U/kg/hour w dextrose D5 or D10 w 20 mEq KCL at 2-3x maintenance	<u>Modifiers</u> - NEpi initiated and quickly titrated → Poor response - Epi and/or Vaso initiated as 2/3rd line agents → Poor response - High-dose insulin and dextrose therapy → Poor response - Request central venous access → Can assume to be placed prn. <u>Triggers</u> - All actions complete or **15 min** into case → **2. Salvage therapy**	
4. Salvage Therapy (180 mins post ingestion) Rhythm: wide complex escape HR: 30 BP: 65/30 on 2-3 vasopressor agents RR: 14 O_2SAT: 90% T: 36.7ºC GCS: 12	*Poorly responsive and confused – improves with ongoing medical therapy*	<u>Expected Learner Actions</u> ☐ Increase high-dose insulin by 1 unit/kg/hour q20-30mins ☐ Repeat ECG with rhythm change to wide complex escape ☐ Discuss lipid emulsion therapy 1.5ml/kg IV over 1-minute repeat 2-3x prn with ICU/Poison Control ☐ Call ICU +/- consideration for ECMO/Transvenous Pacing	<u>Modifiers</u> - Requests ECG → ECG #2 - Escalating high-dose insulin → Improved hemodynamics and mentation (multifactorial) - Lipid emulsion therapy → Discussed with Poison Control and rarely recommended <u>Triggers</u> - Calls ICU for transfer or after 20 minutes total → **End Scenario**	

Appendix A: Laboratory Results

CBC
WBC 9.5
Hgb 140
Plt 420 (H)

Lytes
Na 140
K 3.5
Cl 102
HCO_3 28
AG 7
Urea 5.7
Cr 72
eGFR >120
Glucose 9.0 (H)

Extended Lytes
Ca 1.02 (L)
Mg 0.71
PO_4 1.17
Albumin 45
TSH 2.00

VBG
pH 7.38
pCO_2 35
pO_2 98
HCO_3 28
Lactate 1.4 (H)

Cardiac/Coags
INR 0.9
aPTT 23

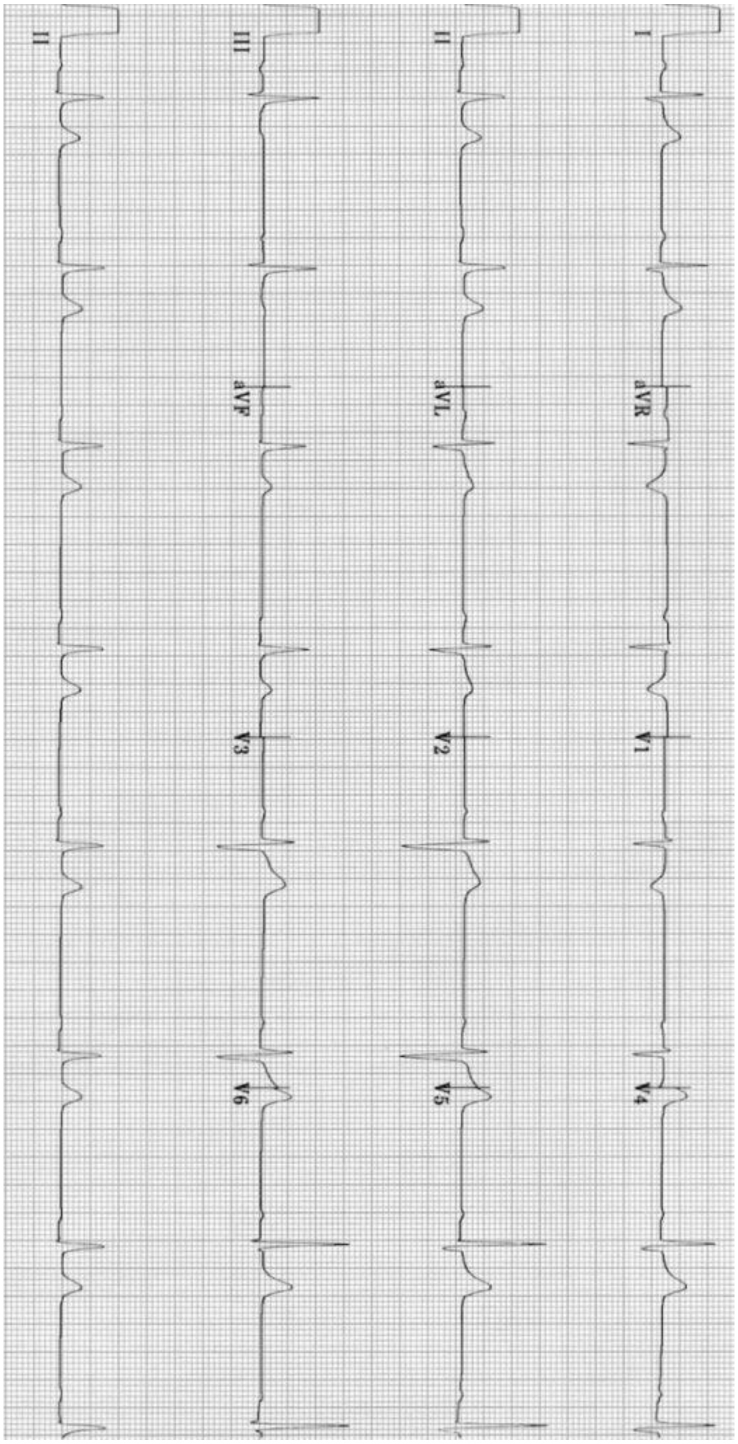
Biliary
AST 27
ALT 23
GGT 13
ALP 69
Bili 12
Lipase 25

Tox
EtOH <2
ASA <0.3
Tylenol <66

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

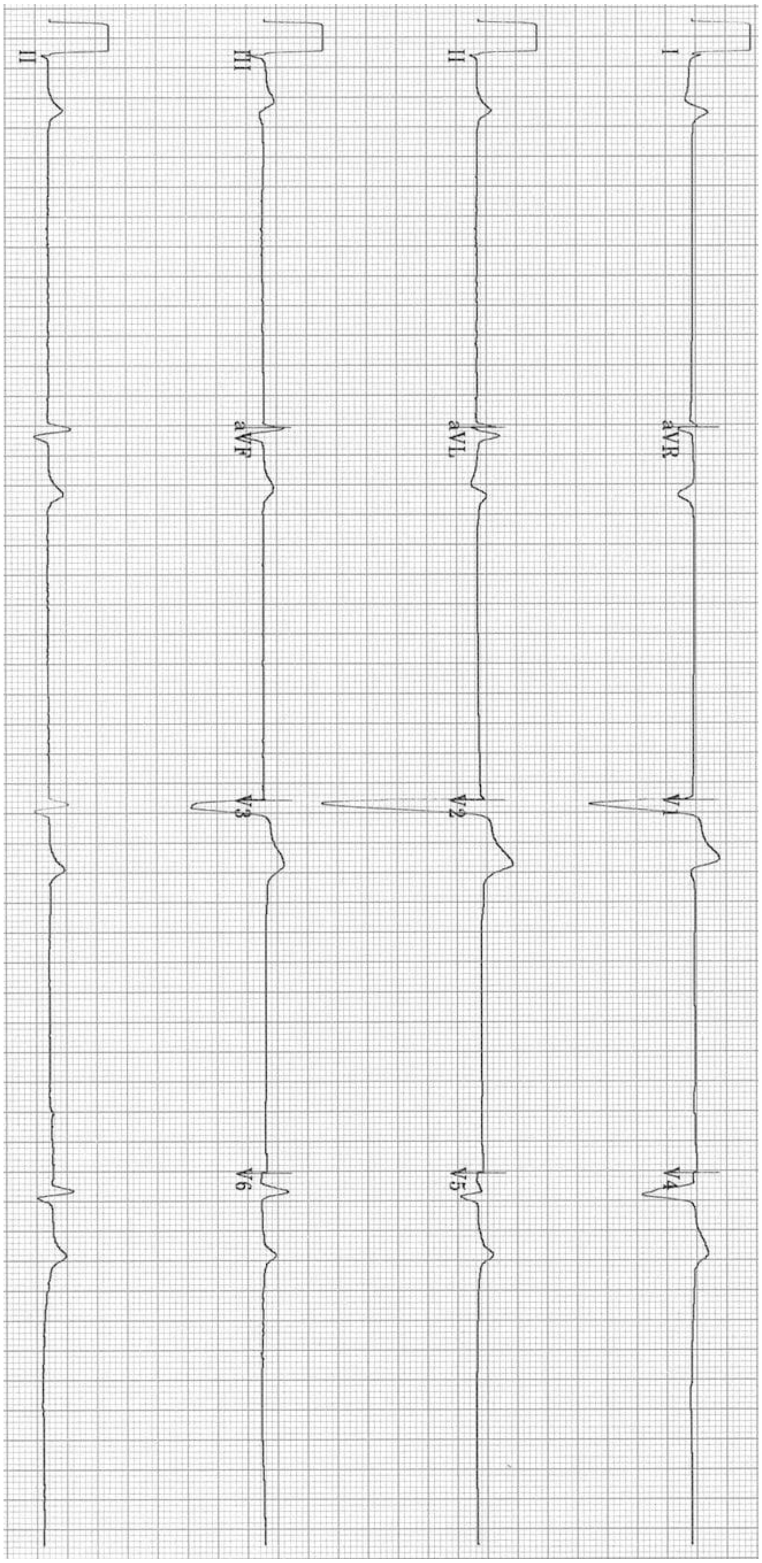
Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!

ECG #1



https://litfl.com/wp-content/uploads/2018/08/ECG-Sinus-bradycardia-1st-degree-AV-block.jpg

ECG #2



https://litfl.com/wp-content/uploads/2018/08/sinus-arrest-ventricular-escape-rhythm-2.jpg

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Debriefing as a group, without video.

Sample questions for debriefing:

CRM

1. How do you prioritize interventions with a patient that is predicted to become critically unwell?
2. How can you use the team to effectively communicate with resources (e.g. poison control/ICU) while managing an unstable patient?

Medical Expert

1. What are the common effects of beta blocker toxicity?
   - Hypotension, bradycardia, AV block, cardiovascular collapse

2. Does it differ with propranolol, acebutolol, and sotalol?
   - **Propranolol and Acebutolol** cause Na channel blockade → QRS widening → $NaHCO_3$ boluses
   - **Sotalol** causes K+ blockade → Long QT → Monitor for Torsades → $MgSO_4$

3. What electrolyte changes can you anticipate in a beta blocker overdose?
   - **Hypokalemia / Hypoglycemia. After starting high dose insulin and dextrose, repeat K+ and Glucose q30 minutes. Hyperglycemia may occur transiently.**

4. What decontamination strategies could be considered in this patient?
   - **Activated charcoal:** Traditionally 1-2 hours from time of ingestion, though most experts advocate for extending the window to 4-6 hours. Administration can be discussed with poison control on a case-by-case basis.
   - **Gastric lavage:** Can be considered as the patient has presented within one hour of presentation. Although rarely recommended, for potentially lethal ingestions, presenting within one-hour, gastric lavage can (and should) be considered. Requires airway protection, thus prophylactic intubation.
   - **Whole bowel irrigation: NOT** currently indicated. May be considered with ingestion of sustained release preparations.

References

1. https://www.bcemergencynetwork.ca/clinical_resource/beta-blockers-overdose/
2. https://www.uptodate.com/contents/beta-blocker-poisoning#H27
3. https://emergencymedicinecases.com/low-slow-poisoning/